

← Back

Voronoi Diagrams and Delaunay triangulations

Graded Quiz - 15 min

≡ Hide menu

Voronoi Diagrams

Voronoi Diagrams: Structure

Complexity of Voronoi Diagrams

Delaunay Triangulations

Angle Optimal Triangulations

Legal Triangulations

Randomized Incremental Construction

Randomized Incremental Construction: Analysis

Programming: Voronoi Diagrams and Delaunay Triangulations

Practice Programming Assignment: Legalizing triangulations 29-40m

Programming Assignment: Legalizing triangulations (Additional inputs) 15 min

Discussion Prompt: Optional programming assignment 1h

Wrap-Up Quiz

Quiz: Voronoi Diagrams and Delaunay triangulations Submitted

🎉 Congratulations! You passed!

Grade received 95.55%

Latest Submission

To pass 80% or better

Retake the assignment in 7h 54m

Go to next item

Voronoi Diagrams and Delaunay triangulations

1. Given that for a set of  $n$  points the number of Voronoi vertices is at most  $2n - 5$  and the number of Voronoi edges is at most  $3n - 6$ , determine the average number of Voronoi vertices of a Voronoi cell? Your answer should be the smallest integer larger than the average. 2 / 2 points

🕒 Correct

coach

Ready to review what you've learned before taking the quiz? I'm here to help.

🔗 Help me practice

💬 Let's chat

📌 Submit your assignment

Due Sep 1, 11:59 PM IST Attempts 3 every 8 hours

Try again

📌 Receive grade

To Pass 80% or higher

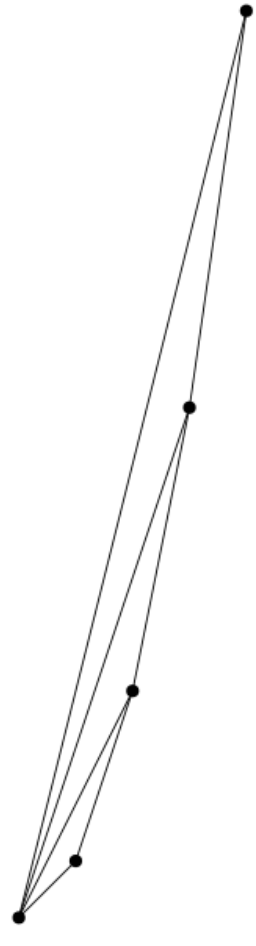
Your grade 95.55%

View Feedback

👍 Like

👎 Dislike

🚩 Report an issue

2. 

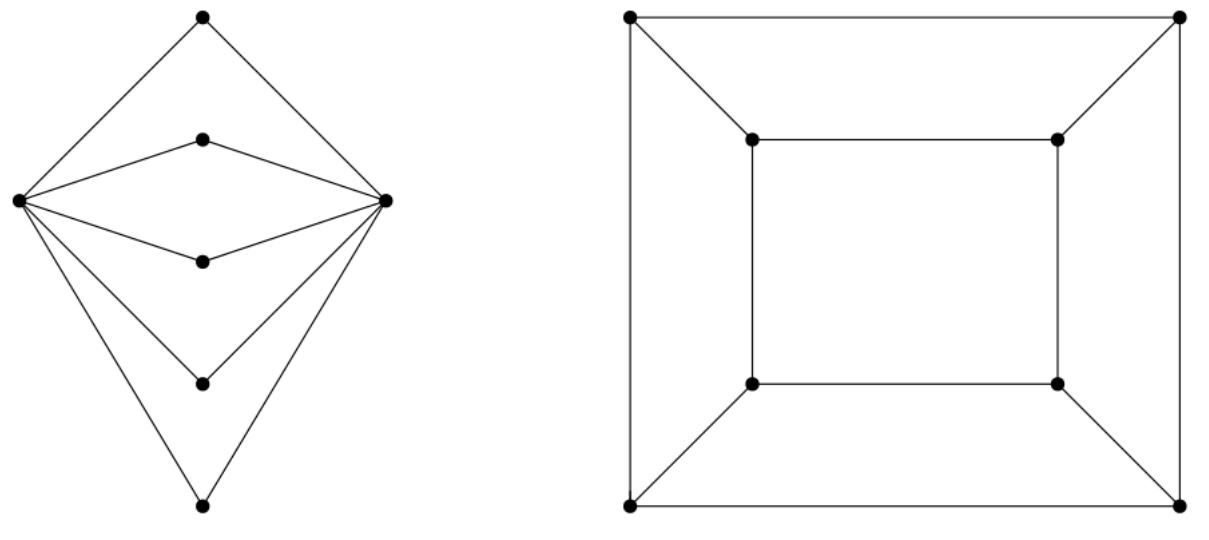
Suppose we use a randomized incremental approach to construct a Delaunay triangulation of the pointset  $\{t_i, t^2\}$  for  $0 \leq i \leq n$ . An example is shown in the image above for  $n = 4$ .

What is the worst order of insertion in terms of running time?

What is the worst case number of triangles created during construction?

What is the expected number of triangles created during construction?

🕒 Incorrect

3. 

Consider connected graphs where every face is a 4-gon, that, is a cycle of four edges. Two examples are shown in the image above. When we know the number of vertices  $|V|$  and the number of edges on the convex hull  $|E_h|$ , we can compute both the number of edges and the number of faces in the graph. Which formulas are correct?

🕒 Correct

https://www.coursera.org/learn/geometric-algorithms/exam/4d7N/Voronoi-diagrams-and-delaunay-triangulations/attempt/redirectToCovertrue

1/1